CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

PROJECT REPORT

Submitted by

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Project Domain

Data Analytics – Banking & Customer Churn

Tools & Technologies

Python | Pandas | Plotly | Streamlit

2. Executive Summary for Government Stakeholders

This project presents a segmentation-driven analysis of customer churn in European banking. The study uses a dataset of 10,000 customer records from France, Germany, and Spain to examine churn patterns across geography, age, credit score, tenure, balance, gender, and customer engagement.

The analysis includes data validation, cleaning, customer segmentation, churn-rate analysis, demographic comparison, high-value customer analysis, and key performance indicators. High-balance churners, salary and balance patterns, and potential revenue exposure from churn are also examined.

An interactive Streamlit dashboard was developed using Python, Pandas, and Plotly. The dashboard provides dynamic filters, KPI updates, visualizations, and customer-level drill-down views. The analysis provides structured, data-driven information that can support customer retention analysis, strategic planning, and evidence-based decision-making.

3. Introduction

Customer churn is an important concern in the banking sector because the loss of existing customers can affect customer relationships, financial performance, and long-term business sustainability. Understanding the characteristics and patterns of customers who leave a bank can help identify customer segments that require greater attention.

This project, titled “Customer Segmentation & Churn Pattern Analytics in European Banking,” focuses on analysing customer churn using exploratory data analysis (EDA) and segmentation-based analysis. The study examines churn patterns across geography, age, gender, credit score, tenure, balance, and customer engagement.

The project transforms customer-level data into meaningful analytical segments and presents the results through an interactive Streamlit dashboard. The overall purpose is to provide a structured view of churn patterns and customer profiles without using a machine-learning prediction model.

4. Problem Statement

Banks may have detailed customer-level information but still face challenges in identifying important churn patterns, comparing churn across regions and demographic groups, and understanding the financial profile of customers who leave. Without structured analytics, churn analysis can remain generic and difficult to explore.

This project addresses the problem through systematic data validation, segmentation, comparative churn analysis, KPI calculations, high-value customer analysis, and interactive dashboard-based drill-down.

5. Objectives

The main objectives of the project are:
1. To analyse customer churn patterns using exploratory data analysis (EDA).
2. To segment customers based on geography, age, credit score, tenure, and balance.
3. To compare churn rates across customer segments and demographic groups.
4. To identify high-value and high-balance churners and examine potential revenue exposure.
5. To develop an interactive Streamlit dashboard with dynamic filters, KPIs, visualizations, and customer-level drill-down analysis.

6. Dataset Description

The dataset contains 10,000 customer records from France, Germany, and Spain. It includes demographic, financial, product, and engagement-related attributes. The Exited field represents churn status, where 1 indicates a churned customer and 0 indicates a retained customer.

The analytical fields used in the project include CustomerId, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited. Surname was excluded from analytical processing because it is not required for segmentation analysis. The Year field present in the source CSV was also excluded because it was outside the defined project analysis scope.

7. Tools & Technologies

• Python — data processing and analytical logic
• Pandas — data cleaning, transformation, grouping, and calculations
• Plotly — interactive visualizations

• Streamlit — interactive web dashboard
• VS Code — development environment
• GitHub — project repository and submission
• Microsoft Word / PDF — project documentation

8. Data Preprocessing & Customer Segmentation

The preprocessing stage included loading the dataset, checking the analytical fields, validating binary variables, removing non-analytical fields, and creating derived segmentation fields.

Customer segmentation was performed across five dimensions:
• Geography: France, Germany, Spain
• Age: <30, 30–45, 46–60, 60+
• Credit Score: Low, Medium, High
• Tenure: New, Mid-term, Long-term
• Balance: Zero-balance, Low-balance, High-balance

The project-defined thresholds were used for credit score, tenure, and balance segmentation. The balance high-value threshold was defined using the 75th percentile of customer balance, representing the top 25% by balance within the analytical dataset.

9. Churn Distribution Analysis

The dashboard calculates overall churn rate, segment-wise churn rates, churn contribution by segment size, and churned-versus-retained customer profiles. The overall churn rate shown in the dashboard is 20.37%.

Geographic churn rates and churn contribution can be explored dynamically. In the unfiltered dataset, Germany showed a higher churn rate than France and Spain, while churn contribution was distributed across all three countries. These measures are presented descriptively and update when filters are applied.

10. Comparative Demographic & Financial Analysis

The analysis compares churn patterns by gender, geography and age, age groups, tenure groups, and financial characteristics. In the unfiltered analysis, female customers had a churn rate of 25.07% and male customers had a churn rate of 16.46%.

Age-based analysis showed churn rates of approximately 7.56% for customers below 30, 15.30% for ages 30–45, 51.12% for ages 46–60, and 24.78% for customers aged 60+. Tenure groups were also compared to identify differences in churn patterns.

Financial analysis examines the relationship between balance, estimated salary, and churn status through comparative visualizations.

11. High-Value Customer Churn Analysis

High-value customer analysis focuses on customers with balances at or above the 75th percentile threshold. The dashboard identifies high-value customers, high-value churners, and the resulting high-value churn rate.

In the unfiltered dashboard view, 2,500 customers were classified as high-value, 592 of these customers were churned, and the high-value churn rate was 23.68%. The project also identifies high-balance churners and examines salary-versus-balance patterns for churned and retained customers.

Potential revenue exposure is analysed descriptively from the financial profiles of churned customers. The analysis does not claim a forecasted financial loss.

12. Key Performance Indicators (KPIs)

The dashboard includes five KPI concepts defined by the project requirements:

1. Overall Churn Rate — percentage of customers who exited.
2. Segment Churn Rate — churn percentage within a selected segment.
3. High-Value Churn Ratio — churn among high-value customers.
4. Geographic Risk Index — regional churn exposure relative to the overall churn rate.
5. Engagement Drop Indicator — difference between churn among inactive and active customers.

In the unfiltered dashboard, the High-Value Churn Ratio is 23.68% and the Engagement Drop Indicator is +12.58 percentage points. KPI values update dynamically when segment filters are changed.

13. Streamlit Dashboard Implementation

The Streamlit application integrates the analytical workflow into an interactive dashboard. Users can filter the analysis by Geography, Age Group, Credit Score, Tenure, and Balance.

The dashboard includes an overall churn summary, geography-wise churn visualization, geographic risk index, churn contribution, churned-versus-retained profile analysis, gender analysis, geography-age comparison, age and tenure comparison, financial stability analysis, salary-versus-balance analysis, high-balance churner analysis, revenue-risk analysis, a high-value customer churn explorer, and customer-level drill-down.

The dashboard was developed using Python, Pandas, Plotly, and Streamlit.

14. Dashboard Evidence



Fig. 1. Streamlit dashboard KPI summary

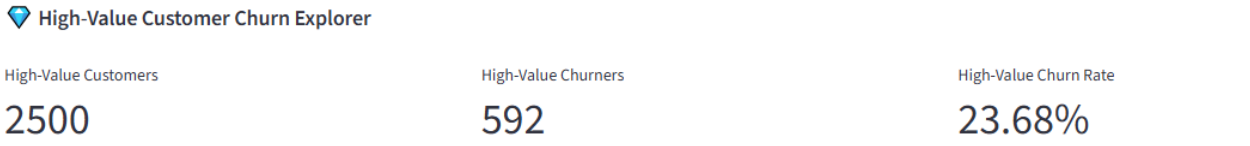


Fig. 2. High-Value Customer Churn Explorer

High-Value Customer Drill-down

CustomerId	Geography	Gender	Age	CreditScore	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Churn Status
15757408	Spain	Male	38	655	3	250898.09	3	0	1	81054	Churned
15715622	France	Female	57	583	3	238387.56	1	0	1	147964.99	Churned
15714241	Spain	Male	42	749	9	222267.63	1	0	0	101108.85	Churned
15586674	Spain	Female	58	663	5	216109.88	1	0	1	74176.71	Churned
15594408	Spain	Female	48	584	2	213146.2	1	1	0	75161.25	Churned
15671256	France	Female	35	850	1	211774.31	1	1	0	188574.12	Churned
15736420	France	Male	21	596	4	210433.08	2	0	1	197297.77	Churned
15721658	Spain	Female	56	672	2	209767.31	2	1	1	150694.42	Churned
15578671	Spain	Female	29	706	1	209490.21	1	1	1	133267.69	Churned
15709920	France	Female	33	479	2	208165.53	1	0	0	50774.81	Churned

Fig. 3. High-Value Customer Drill-down

15. Results & Key Insights

The analysis demonstrates that churn is not uniform across customer segments. Geographic, age, gender, tenure, credit score, balance, and engagement characteristics show measurable differences in churn patterns.

The high-value customer analysis shows that churn can occur among customers with comparatively high account balances. The dashboard makes these patterns explorable through dynamic filtering rather than presenting only a fixed summary.

The customer drill-down view provides customer-level details for high-value churners, including geography, gender, age, credit score, tenure, balance, product count, card ownership, activity status, estimated salary, and churn status.

16. Limitations

• The analysis is based on the available customer dataset and its recorded attributes.
• The project is descriptive and exploratory; it does not implement a machine-learning churn prediction model.
• Credit-score, tenure, and balance bands are project-defined analytical segments because the requirements do not specify exact thresholds.
• The Geographic Risk Index and Engagement Drop Indicator are analytical measures defined for this dashboard.
• Revenue exposure is analysed descriptively from customer financial profiles and should not be interpreted as a forecast of actual future financial loss.

17. Future Scope

Future development could extend the dashboard with additional time-based analysis if suitable dated customer activity data becomes available. Predictive churn modelling, customer lifetime value analysis, automated alerts, advanced cohort analysis, and integration with live banking data could also be considered in a future version.

Any future predictive component should be validated separately using appropriate model evaluation, data governance, privacy, and responsible-use practices.

18. Conclusion

This project presents a segmentation-driven exploratory analysis of customer churn in European banking. The analysis examines churn across geography, age, gender, credit score, tenure, balance, and customer engagement.

Customer segmentation and comparative analysis provide a structured way to understand differences between churned and retained customers. High-value customer analysis further highlights churn among customers with higher account balances and provides a descriptive view of potential financial exposure.

The Streamlit dashboard integrates these analyses through dynamic filters, KPI updates, interactive visualizations, and customer-level drill-down. Overall, the project demonstrates how exploratory data analysis and customer segmentation can transform customer-level banking data into structured analytical insights.

19. References

[1] A. Agarwal, S. Prabha, and R. Yadav, “Exploratory Data Analysis (EDA) for Banking and Finance: Unveiling Insights and Patterns,” GBS Impact: Journal of Multi Disciplinary Research, vol. 11, no. 1, pp. 249–280, 2025. DOI: 10.58419/gbs.v11i1.1112520

[2] G. Kaur, R. D. Sharma, and N. Mahajan, “Segmentation of Bank Customers by Loyalty and Switching Intentions,” Vikalpa: The Journal for Decision Makers, vol. 39, no. 4, pp. 75–90, 2014. DOI: 10.1177/0256090920140406

Appendix A. Customer Segmentation Definitions

Dimension	Segments	Project Definition
Geography	France; Germany; Spain	Dataset categories
Age	<30; 30–45; 46–60; 60+	Age bands
Credit Score	Low; Medium; High	Low <580; Medium 580–669; High ≥670
Tenure	New; Mid-term; Long-term	New 0–2; Mid-term 3–7; Long-term 8–10
Balance	Zero; Low; High	Zero = 0; Low >0 to median; High > median